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Subject: Programming with Java

Sem: III AY: 2026-27

Assignment No:

TITLE: Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives

To demonstrate the use of the super keyword for accessing parent class members.

Theory

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.

Exercise

1. Create an Employee and Manager program where Manager accesses Employee details using super keyword.

Users > sumit > Documents > java > java1 > EmployeeManagerDemo.java > Manager > Manager()

```
1  class Employee
2  {
3      String name = "Sumit";
4      double salary = 50000;
5
6      Employee()
7      {
8          System.out.println("Employee Constr");
9      }
10
11     void display()
12     {
13         System.out.println("Employee Name: " + name);
14         System.out.println("Employee Salary: " + salary);
15     }
16 }
17
18 class Manager extends Employee
19 {
20     String name = "Karan";
21
22     Manager()
23     {
24         super();
25         System.out.println("Manager Constr");
26     }
27
28     void display()
29     {
30         System.out.println("Manager Name: " + name);
31         System.out.println("Employee Name: " + super.name);
32         System.out.println("Employee Salary: " + super.salary);
33         super.display();
34     }
35 }
36
37 public class EmployeeManagerDemo
38 {
39     Run | Debug
40     public static void main(String[] args)
41     {
42         Manager m = new Manager();
43         m.display();
44     }
```

Output:

```
[Running] cd "/Users/sumit/Documents/java/java1/" && javac EmployeeManagerDemo.java && java EmployeeManagerDemo
Employee Constr
Manager Constr
Manager Name: Karan
Employee Name: Sumit
Employee Salary: 50000.0
Employee Name: Sumit
Employee Salary: 50000.0
```

2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.

```
Users > sumit > Documents > java > java1 > VehicleInsuranceDemo.java > VehicleInsuranceDemo
1  class Vehicle
2  {
3      String vehicleName = "Car";
4      String vehicleNumber = "MH12AB1234";
5
6      Vehicle()
7      {
8          System.out.println("Vehicle Constr");
9      }
10
11     void display()
12     {
13         System.out.println("Vehicle Name: " + vehicleName);
14         System.out.println("Vehicle Number: " + vehicleNumber);
15     }
16 }
17
18 class CarInsurance extends Vehicle
19 {
20     String insuranceType = "Comprehensive";
21 }
```

```

22     CarInsurance()
23     {
24         super();
25         System.out.println("Car Insurance Constr");
26     }
27
28     void display()
29     {
30         System.out.println("Insurance Type: " + insuranceType);
31         System.out.println("Vehicle Name: " + super.vehicleName);
32         System.out.println("Vehicle Number: " + super.vehicleNumber);
33         super.display();
34     }
35 }
36
37 public class VehicleInsuranceDemo
38 {
39     Run | Debug
40     public static void main(String[] args)
41     {
42         CarInsurance c = new CarInsurance();
43         c.display();
44     }

```

Output:

```

[Running] cd "/Users/sumit/Documents/java/java1/" && javac VehicleInsuranceDemo.java && java VehicleInsuranceDemo
Vehicle Constr
Car Insurance Constr
Insurance Type: Comprehensive
Vehicle Name: Car
Vehicle Number: MH12AB1234
Vehicle Name: Car
Vehicle Number: MH12AB1234

[Done] exited with code=0 in 0.373 seconds

```

Conclusion:

The programs successfully demonstrated the use of the super keyword in Java through real-world examples. The Employee and Manager program showed how a child class can access the parent class variables, constructor, and methods. The Vehicle Insurance System demonstrated similar access to parent vehicle information. The super() keyword was used to call the parent constructor, while super.variable and super.method() accessed parent members. Thus, the super keyword is useful for accessing and reusing parent class members in inheritance.